MotionIQ: End-to-End Video Processing Pipeline & Dataset Feature Catalog

Comprehensive Technical Documentation of Workflow Steps, 23 Dataset Features, Mathematical Formulas & ML Usage

1. Step-by-Step Video Upload & Analysis Workflow

When an optical movement video is uploaded to MotionIQ (or recorded via live webcam), it undergoes a 6-stage sequential processing workflow transforming raw RGB video frames into 3D skeletal kinematics and ML injury predictions:

• Step 1: Video Ingestion & OpenCV Frame Extraction

Extracts optical video frames at 60 FPS. Validates file resolution, aspect ratio, and frame integrity.

• Step 2: Google MediaPipe 33-Keypoint Pose Tracking

Locates 33 3D skeletal landmarks per frame: P = (x, y, z, visibility), where x, y in [0, 1] represent normalized spatial coordinates.

• Step 3: Biomechanical Vector Geometry & Joint Formulas

Calculates frame-by-frame joint angles (Knee, Hip, Ankle, Elbow), Range of Motion (ROM), and Left/Right bilateral symmetry percentage.

• Step 4: Posture Alignment & Kinematic Fatigue Index

Measures forward trunk lean angle relative to vertical axis (0 deg) and derives Kinematic Fatigue Index from frame-to-frame velocity variability.

• Step 5: Supervised ML Inference (XGBoost & Random Forest)

Compiles derived features and athlete workload metadata into a 13D feature vector passed into trained ML models to compute Injury Risk %.

• Step 6: Clinical PDF Generation & Real-time Dashboard Update

Generates downloadable multi-page PDF reports, prescribes targeted physical therapy exercises, and renders 3D skeletal heatmaps on UI.

2. Complete Catalog of 23 Features (Dataset Source, Biomechanical Purpose & ML Usage)

Below is the complete, detailed catalog of all 23 features used across our datasets and video analysis engine. Each entry specifies the feature name, source dataset, clinical reason for use, and exact role in ML inference:

Feature Name	Dataset Source	Clinical / Biomechanical Reason	How it is Used in ML Models
1. Knee_Angle_deg / knee_flexion	Dataset 2 (Project-Injury) & MediaPipe Video	Primary metric for squat depth, deceleration landing safety, and knee joint flexion capacity.	Derived via acos vector dot product at Hip-Knee-Ankle. Input to ML injury category model and risk penalty formula.
2. Ankle_Flexion_deg / ankle_dorsiflexion	Dataset 2 (Project-Injury) & MediaPipe Video	Restricted ankle dorsiflexion forces kinetic ground impact shock upward into knee shearing forces.	Derived via acos vector dot product at Knee-Ankle-Foot. Input to ML injury classifier and mobility assessment.
3. range_of_motion (ROM)	Dataset 1 (sports_multimodal) & MediaPipe Video	Measures total angular excursion across frames: max(angle) - min(angle). Indicates mobility vs joint stiffness.	Feature vector element #1 in XGBoost Model 1. Used to calculate bilateral leg symmetry score.
4. gait_symmetry / symmetry_score	Dataset 1 (sports_multimodal) & MediaPipe Video	Identifies Left vs Right leg imbalance, favoring/limping, and unilateral load overload.	Calculated as 100 * (1 - |ROM_L - ROM_R| / max(ROM_L, ROM_R)). Feature vector element #2 in Model 1.
5. body_orientation / trunk_lean_deg	Dataset 1 (sports_multimodal) & MediaPipe Video	Measures forward/lateral spinal postural tilt. Excessive lean increases lower back strain & ACL shear.	Derived via shoulder-hip vector relative to vertical (0, -1). Feature vector element #3 in Model 1.

Feature Name	Dataset Source	Clinical / Biomechanical Reason	How it is Used in ML Models
6. fatigue_index / Fatigue_Score	Dataset 1 & 3 & MediaPipe Video	Neuromuscular fatigue degrades joint control and increases frame-to-frame movement tremor.	Derived as 100 - Consistency %. Feature vector element #4 in Model 1 & Model 4. Triggers alert if > 25.0 pts.
7. previous_injury_history	Dataset 1 (sports_multimodal_data)	Prior ACL, meniscus, or hamstring injury is the single highest statistical predictor of re-injury.	Binary encoded (0 = No, 1 = Yes). Feature vector element #5 in Model 1. Adds +15.0 pts risk penalty.
8. repetition_count	Dataset 1 (sports_multimodal_data)	Higher repetition counts increase cumulative tissue loading and progressive fatigue degradation.	Numeric count of completed exercise reps. Feature vector element #6 in Model 1.
9. workload_intensity / Training_Intensity	Dataset 1 & Dataset 3 (collegiate_athlete)	Session exertion rating; high workload intensity without rest drives micro-trauma overuse.	Rated 1 - 10. Feature vector element #7 in Model 1 and Model 4 (ACL Regressor).
10. ground_reaction_force	Dataset 1 (sports_multimodal_data)	Quantifies kinetic force exerted by the ground on lower limb joints upon landing (Newtons).	Force in Newtons. Feature vector element #8 in XGBoost Model 1.
11. impact_force	Dataset 1 (sports_multimodal_data)	Measures peak transient shock load absorbed by cartilage and ligaments upon ground contact.	Peak impact in Newtons. Feature vector element #9 in XGBoost Model 1.
12. angular_velocity	Dataset 1 (sports_multimodal_data)	Speed of joint rotation (deg/sec); rapid uncontrolled flexion causes acute ligament tearing.	Angular speed in deg/s. Feature vector element #10 in XGBoost Model 1.
13. acceleration	Dataset 1 (sports_multimodal_data)	Rate of body velocity change (m/s^2); rapid deceleration forces trigger non-contact ACL stress.	Acceleration in m/s^2. Feature vector element #11 in XGBoost Model 1.
14. jump_height / Jump_Height_cm	Dataset 1 & Dataset 2 (Project-Injury)	Measures lower-body explosive power and vertical landing kinetic energy.	Height in meters/cm. Feature vector element #12 in Model 1 & Model 2.
15. speed / Speed_m_s	Dataset 1 & Dataset 2 (Project-Injury)	Movement velocity; higher speeds generate higher landing kinetic energy and torque.	Velocity in m/s. Feature vector element #13 in Model 1 & Model 2.
16. Age	Dataset 2 & Dataset 3 (collegiate_athlete)	Age influences tissue elasticity, collagen recovery rate, and baseline ligament vulnerability.	Age in years. Input feature in ML Models 2, 3, and 4.
17. Height_cm	Dataset 2 & Dataset 3 (collegiate_athlete)	Taller stature increases joint moment arms and mechanical lever torque on knees and hips.	Height in centimeters. Input feature in ML Models 2, 3, and 4.
18. Weight_kg	Dataset 2 & Dataset 3 (collegiate_athlete)	Higher body mass increases ground reaction impact force and joint compressive load.	Body mass in kilograms. Input feature in ML Models 2, 3, and 4.
19. Reaction_Time_ms	Dataset 2 (Project-Injury-Dataset)	Slower reaction time leads to delayed muscle activation during unexpected land/cut forces.	Latency in milliseconds. Input feature in Models 2 and 3.
20. Sport_Encoded	Dataset 2 (Project-Injury-Dataset)	Different sports (Football vs Basketball vs Track) have distinct injury incidence patterns.	LabelEncoded integer (0 to K). Input feature in Models 2 and 3.
21. Training_Hours_Per_Week	Dataset 3 (collegiate_athlete_injury)	Accumulated weekly training volume; excessive volume without rest leads to overuse injuries.	Weekly hours. Input feature in Model 4 (Continuous ACL Risk Regressor).
22. Recovery_Days_Per_Week	Dataset 3 (collegiate_athlete_injury)	Days per week dedicated to rest/recovery; critical for muscle tissue repair and collagen synthesis.	Rest days count (0 to 4). Input feature in Model 4 (Continuous ACL Risk Regressor).

Feature Name	Dataset Source	Clinical / Biomechanical Reason	How it is Used in ML Models
23. Dynamic_Knee_Valgus_deg	MediaPipe Video & Biomechanical Engine	Inward knee collapse during deceleration landings; #1 mechanical cause of non-contact ACL tears.	Evaluated via frontal-plane knee-to-ankle medial displacement. Triggers high-risk ACL alert.

3. Summary of Machine Learning Models & Feature Arrays

MotionIQ combines the 23 features above into 4 specialized Machine Learning models:

1. **Model 1 (Kinematic Injury Risk Classifier)**: XGBoost / Random Forest on 13 features (ROC-AUC = 0.941).
2. **Model 2 (Specific Injury Type Classifier)**: Multi-class classifier on 9 features (Accuracy = 92.4%).
3. **Model 3 (Rehabilitation Prescriptor)**: Prescribes rehab program and recovery duration in weeks.
4. **Model 4 (Continuous ACL Risk Regressor)**: Predicts continuous ACL strain score on 8 workload features.